

Data science leader with 10+ years of experience building and scaling machine learning systems in enterprise environments. Proven track record leading cross-functional teams, deploying production-grade ML/AI platforms, and translating complex data into measurable business outcomes across risk, fraud, and governance-heavy domains. Recently led enterprise data science integration strategy during the Capital One–Discover merger, aligning platforms, workflows, and model governance across organizations.

Expertise

- Strategic Leadership & Governance:** ML/AI Strategy, MLOps Governance, Data Governance, Model Risk Management, Technical Leadership, Organizational Change Management, Enterprise Integration.
- AI & Advanced Modeling:** Predictive Modeling, Natural Language Processing, Graph Analytics, Representation Learning, Vector Search, Agentic RAG, Generative AI.
- Data Engineering & Infrastructure:** Distributed Systems, Cloud Architecture, ETL/ELT, MLOps, Model Deployment, Production ML, LLMOps.
- Decision Sciences:** Risk Analytics, Fraud Detection, Experimentation, Optimization, Simulation, Stochastic Modeling.

Recent Professional Experience

Capital One (Formerly: Discover Financial Services) January 2023 – April 2026

Manager, Data Science | February 2025 – April 2026

- Defined and executed enterprise data science integration strategy following the Capital One–Discover merger, aligning analytics platforms, ML workflows, and governance standards across organizations.
- Directed harmonization of complex data structures, ETL pipelines, and analytical infrastructure spanning previously independent technology ecosystems.
- Partnered with senior executives to prioritize integration roadmaps, align technical investments with strategic objectives, and ensure continuity of critical data science capabilities.

Key Accomplishments:

- Drove adoption of unified ML/AI frameworks with MLOps and LLMOps best practices, incorporating Model Context Protocols (MCP) for seamless tooling integrations, reducing model deployment friction by 33% and improving governance consistency across enterprise production systems.
- Managed a 5-person data science team through large-scale organizational transformation, maintaining delivery velocity, operational stability, and team engagement during post-merger integration.
- Developed migration strategies for proprietary ML platforms and advanced analytics tooling, minimizing business disruption while accelerating platform unification.

Lead Data Scientist | January 2023 – February 2025

- Led a cross-functional data science team focused on representation learning, relational network analysis, and predictive modeling for enterprise risk, audit, and compliance functions.
- Architected advanced graph and embedding frameworks to uncover latent relationships across complex financial and operational datasets.
- Developed enterprise reporting and analytics frameworks for audit performance, enabling automated reporting, executive visibility, and real-time operational monitoring.

Key Accomplishments:

- Built scalable machine learning pipelines for fraud detection and risk scoring, improving model performance while reducing false positives across high-volume financial workflows.
- Designed and deployed an end-to-end Issue Analysis platform leveraging NLP, agentic AI, graph analytics, and predictive modeling to proactively identify compliance risks and operational anomalies across multiple audit domains.
- Established engineering and MLOps best practices, developing a departmental training program and mentoring 12 junior data scientists in model development, feature engineering, testing, and production deployment.

United States Army September 2015 – September 2019

Budget Analyst / Resource Manager — Mission Command Center of Excellence

- Built analytical dashboards supporting budget forecasting, scenario planning, and executive decision-making.

Compliance Officer — Combined Arms Command G8

- Oversaw internal financial controls and regulatory compliance, strengthening audit readiness and ensuring adherence to Department of Defense fiscal regulations.

Logistics Specialist — 40TH Military Police Battalion S4

- Coordinated procurement, supply chain operations, and asset readiness across mission-critical logistics functions.

Key Accomplishments:

- Managed multiple operational budgets, each exceeding \$3.75M, aligning financial execution with strategic priorities while improving forecasting accuracy and resource allocation efficiency.
- Improved review processes by integrating analytical pattern-detection methodologies that reduced audit cycle times and increased control effectiveness.
- Automated asset tracking and logistics workflows, reducing administrative overhead while improving operational readiness.
- Recruited and fully trained replacement personnel ahead of scheduled transition timelines, ensuring uninterrupted mission continuity.

Additional Professional Experience

Polysophy Inc.*Founder, Executive Director* | August 2011 – September 2015

New York City College of Technology*Research Associate* | January 2009 – November 2014

Education & Certificates

Master of Science, Analytics – Computational Data Analytics | Georgia Institute of Technology

Bachelor of Science, Applied Mathematics – Finance Concentration | New York City College of Technology

Certified ScrumMaster (CSM) | Scrum Alliance

Plan, Program, Budget, and Execution System (PPBE) | U.S. Army Financial Management School

Technologies

Core: Python, SQL, Spark, Databricks, AWS (SageMaker), Azure ML, Docker, Kubernetes

Machine Learning & AI: LangChain, LangGraph, TensorFlow, Hugging Face, Neo4j, NetworkX, spaCy, NLTK, Gensim

Analytics & Visualization: Tableau, Power BI, Looker, Plotly, Dash, Alteryx

Operations Research & Simulation: Arena, SimPy, ERP Systems